

Lab 33 Worksheet — File Systems, Permissions and Share Configuration

Name: ____ Date: __

Exam objective: Compare NTFS, FAT32 and exFAT, and configure NTFS and share permissions correctly, applying the most-restrictive rule (Core 2 objectives 1.9 and 2.5).

Record as you go

Step	What you did	What you observed
1	Build the comparison table for NTFS, FAT32 and exFAT with columns for maximum file size, maximum volume size, permissions support, encryption, journaling and compatibility.	
2	Record the decisive limits: FAT32 cannot hold a file larger than 4 GB, exFAT removes that limit but has no permissions, and NTFS supports permissions, encryption, compression and journaling.	
3	Record which file system to choose for a USB drive shared with macOS, an internal Windows system drive, and a drive holding files larger than 4 GB.	
4	Confirm the file system on your own volumes.	
5	Create a test folder structure for the permissions exercise.	
6	Display the current NTFS permissions and identify which are inherited from the parent folder.	
7	Grant a specific user read-only NTFS permission on the folder and confirm the change.	
8	Record the standard NTFS permission levels in increasing order: Read, Read and Execute, List	

Step	What you did	What you observed
	Folder Contents, Write, Modify and Full Control.	
9	Explain inheritance: child files and folders inherit permissions from the parent unless inheritance is explicitly broken, and record why breaking inheritance must be deliberate.	
10	Share the folder over the network and set a share permission that differs from the NTFS permission.	
11	State the effective permission for a user with Full Control at the share level and Read at the NTFS level, and confirm the most-restrictive rule gives Read.	
12	Record that NTFS permissions apply both locally and over the network while share permissions apply only over the network, then remove the share and clean up.	

Verification

Success criterion: Your comparison table states the 4 GB FAT32 file size limit and identifies which file systems support permissions; icacls confirms the granted permission; and you correctly state that Share Full Control combined with NTFS Read yields an effective permission of Read.

- I completed every step in the lab.
- My result meets the success criterion above.
- I recorded my evidence (screenshots, output, completed tables).

Reflection

What surprised you in this lab?

Where would you apply this on the job?

What do you still need to revise before the exam?
